

Converse bounds for quantum hypothesis exclusion: A divergence-radius approach

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(also see companion paper: arXiv:2407.13728)

Hypothesis exclusion

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- Similar to hypothesis testing, a finite set of hypotheses is given, say, $\mathcal{X} \equiv \{1, 2, \dots, |\mathcal{X}|\}$
 - One hypothesis is true (randomly selected according to a prior probability distribution), others are false
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- Unlike hypothesis testing (where the experimenter is asked to guess the true hypothesis), here the experimenter is asked to **rule out one false hypothesis**
- Loses if the experimenter rules out the true hypothesis (unsuccessful exclusion), otherwise wins
 - Successful exclusion is much easier to accomplish than successful testing

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- An example
 - 3 suspects: 1 guilty, 2 innocent
 - Range of each suspect's height is known

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suspect 1	suspect 2	suspect 3
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 - (Never know for sure who is guilty, but always know for sure a certain suspect is innocent)
- Practical motivation
 - Ruling out a wrong answer is a meaningful baby step towards determining the right answer
 - Simple multi-hypotheses generalisation of binary symmetric hypothesis testing
 - Quantum state exclusion has applications in foundations of quantum theory, e.g., ontological interpretations of quantum states (Pusey-Barrett-Rudolph, 2012)

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- Expect the error probability to decay exponentially: $P_{\text{err}}(\mathcal{E}^n) \sim e^{-cn}$
 - define the asymptotic error exponent as

$$E_{\text{err}}(\mathcal{E}) := \liminf_{n \rightarrow +\infty} -\frac{1}{n} \ln P_{\text{err}}(\mathcal{E}^n)$$

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where

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Best efficiently computable converse bound known so far for quantum state exclusion

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The converse bound is thus tight (achievable) for **classical** state exclusion (Mishra et al., 2024)

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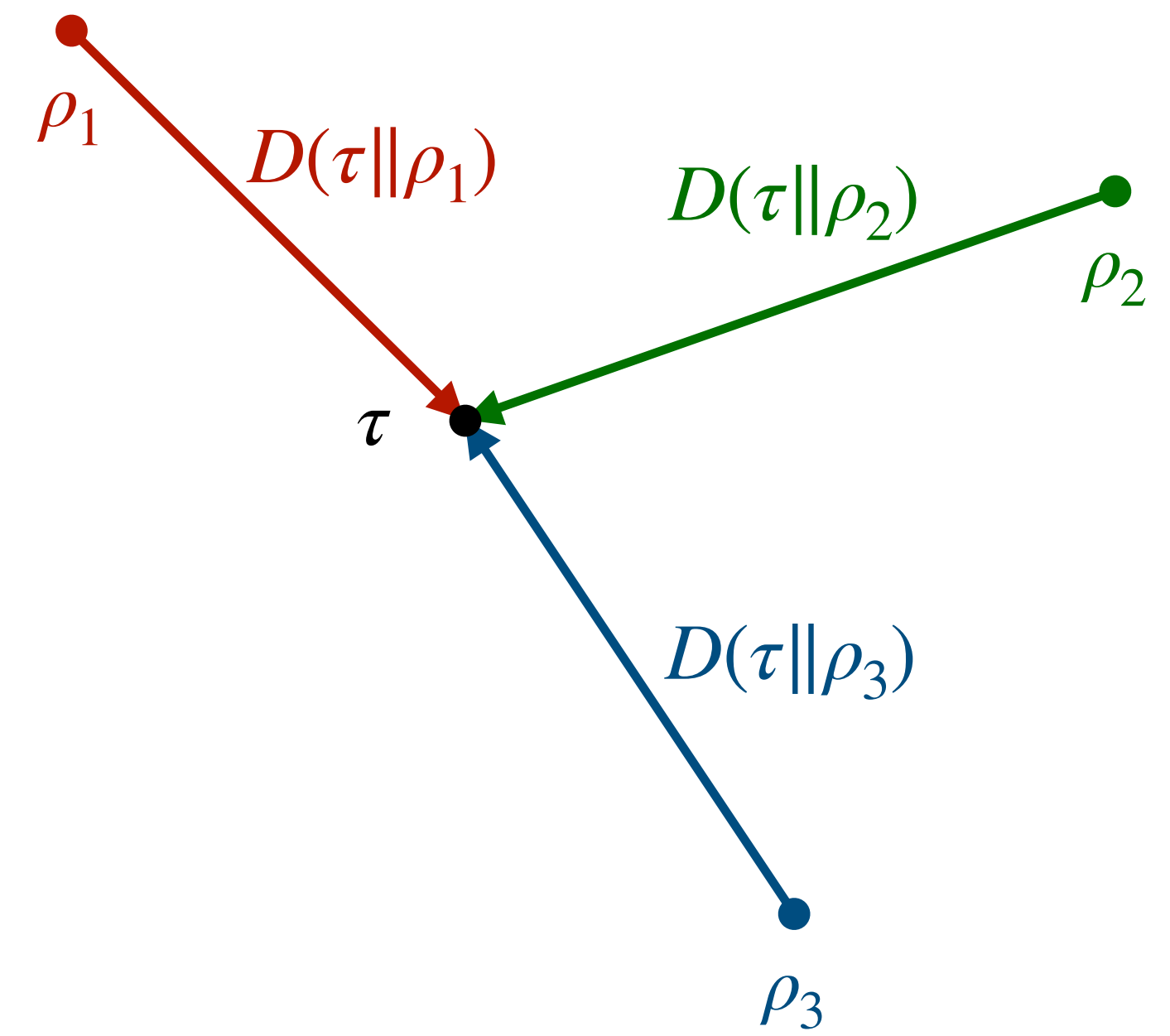
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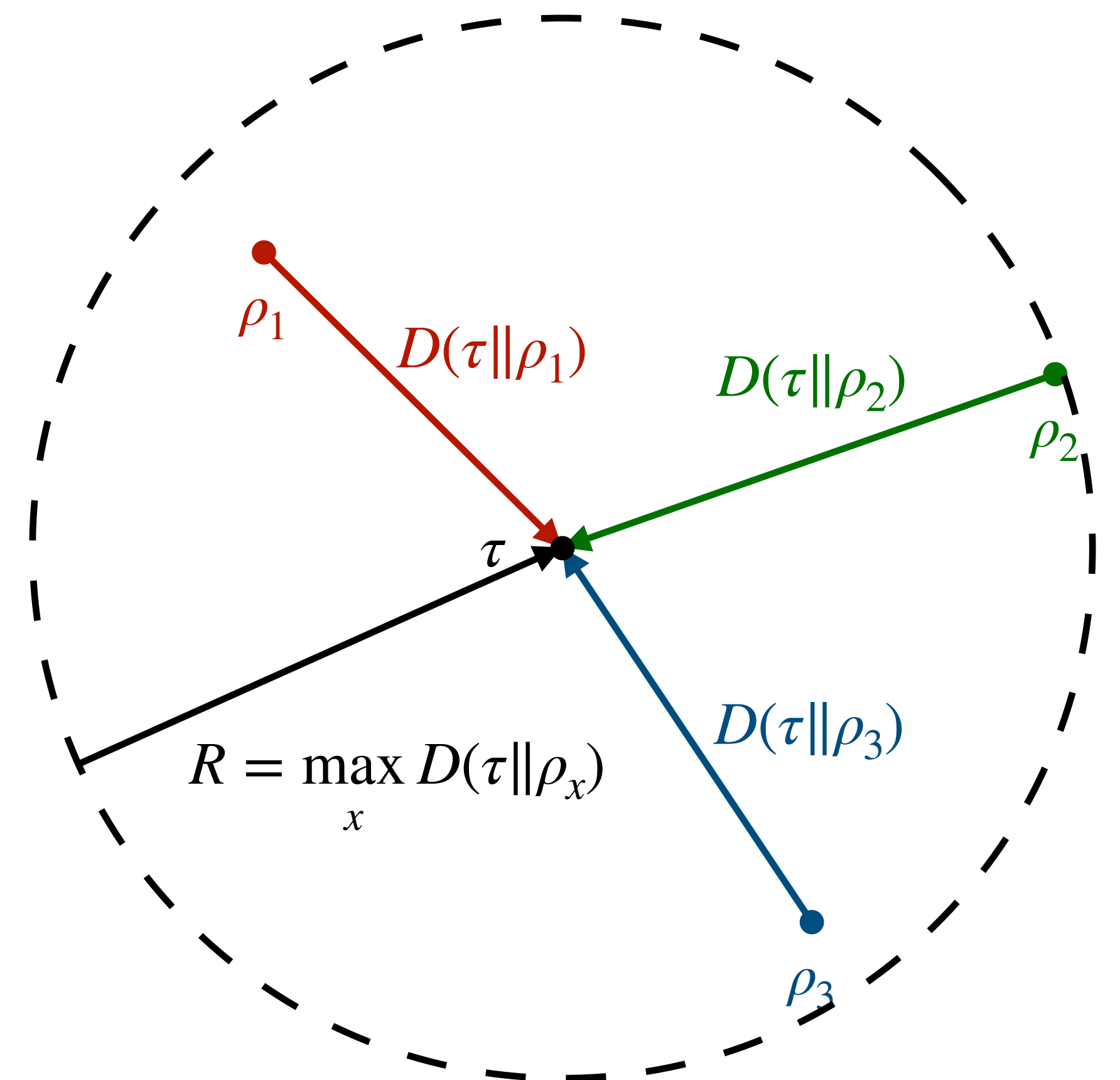
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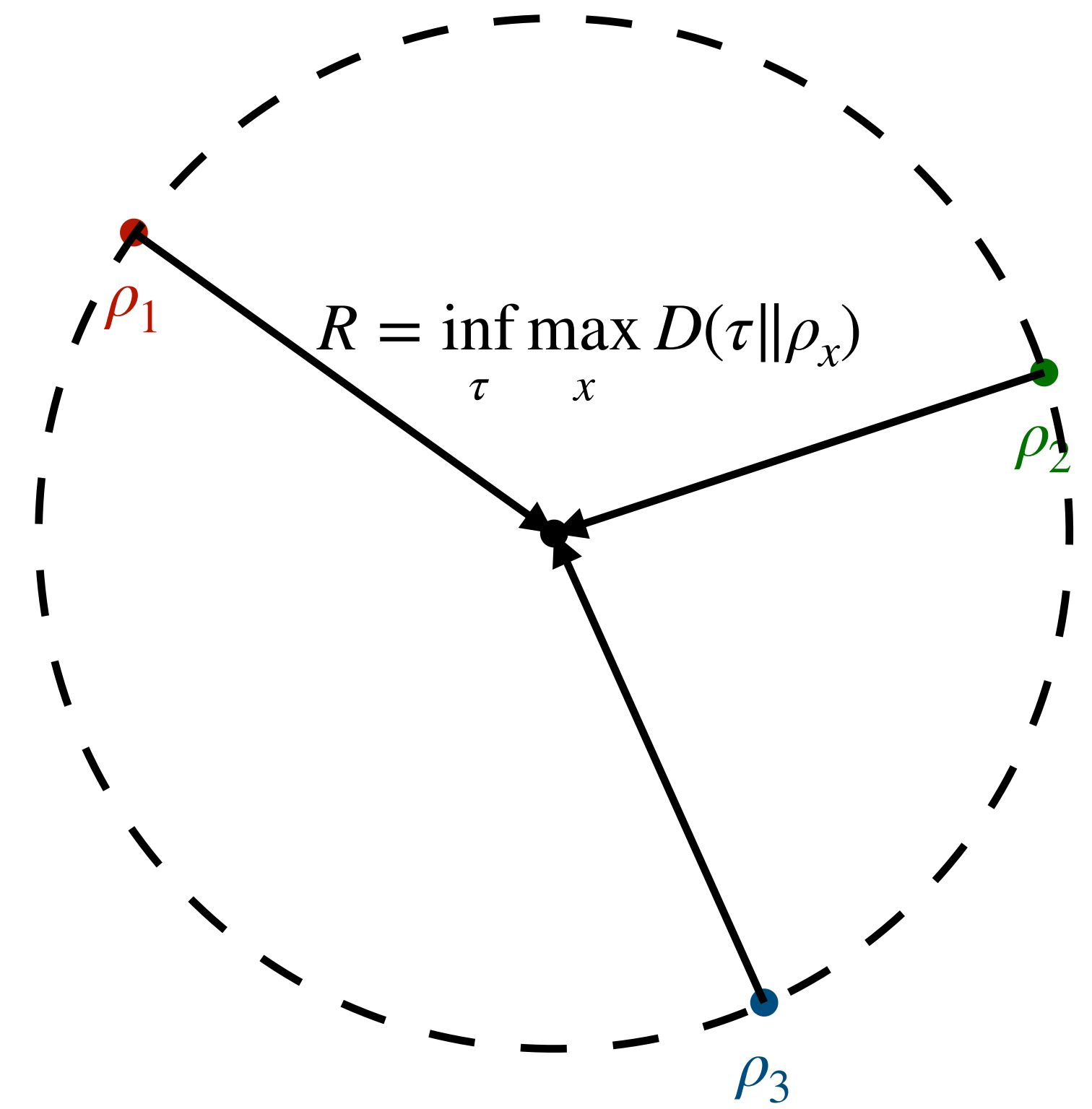
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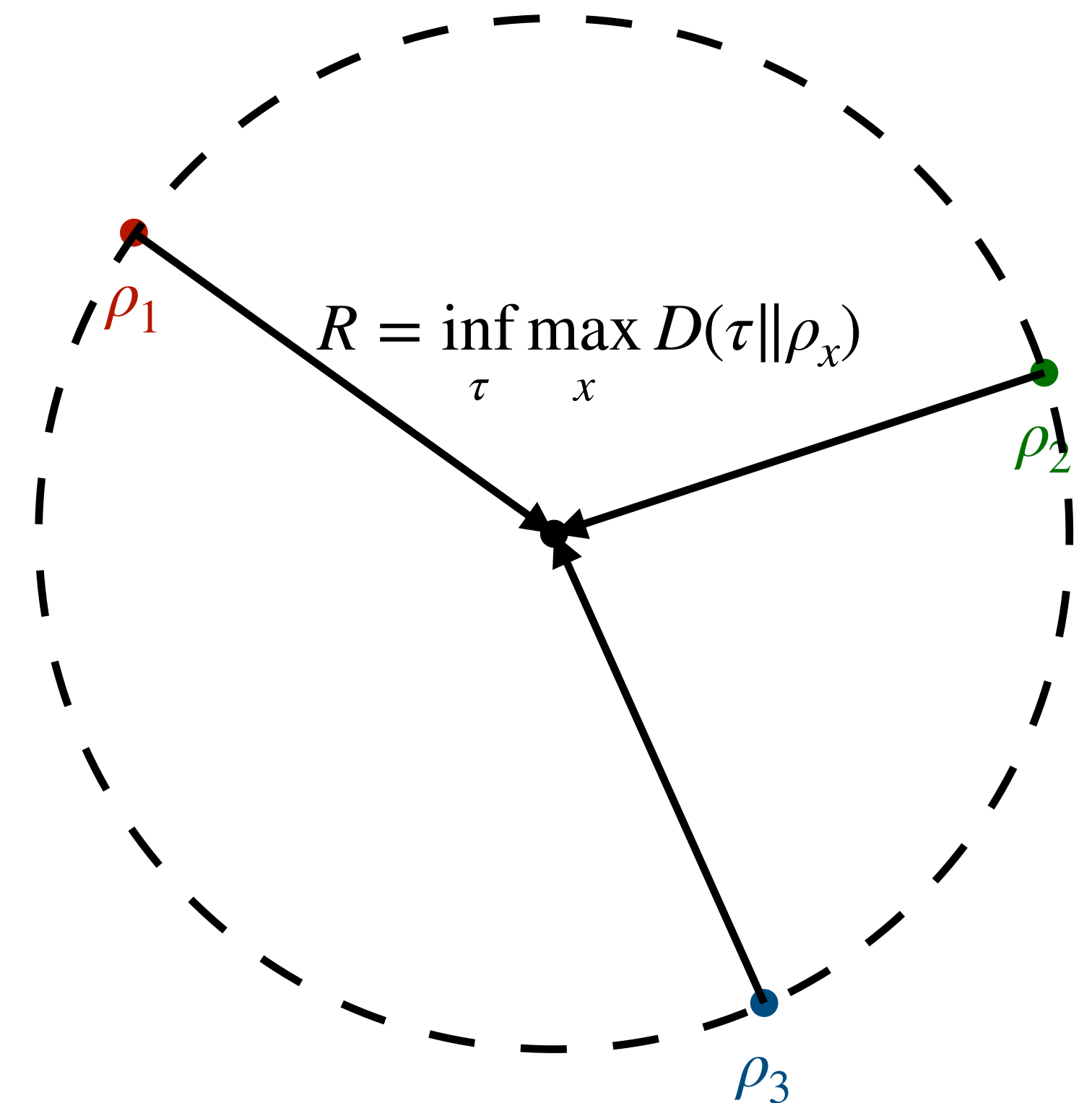
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radius of the smallest sphere that contains all the states, given that the “distance” from the centre to each state is measured by a given divergence



Graphical proof of the converse bound

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(Nagaoka and Ogawa, 2000)

Quantum Stein's lemma, strong converse:

Let τ and ρ be two quantum states and $(\Lambda^{(n)})_n$ be an infinite sequence of measurement operators, with $0 \leq \Lambda^{(n)} \leq I^{(n)}$ for all n . If

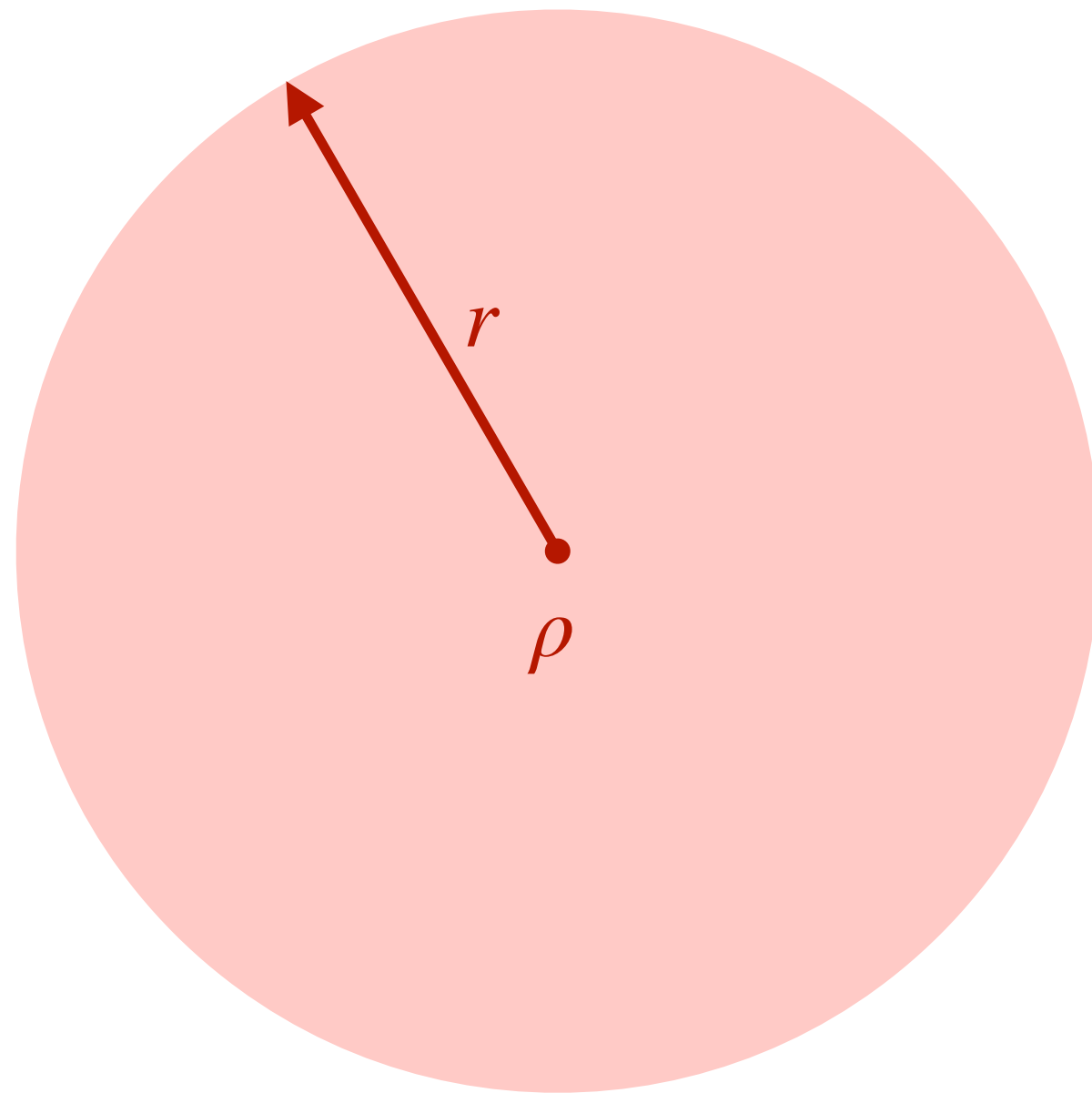
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then

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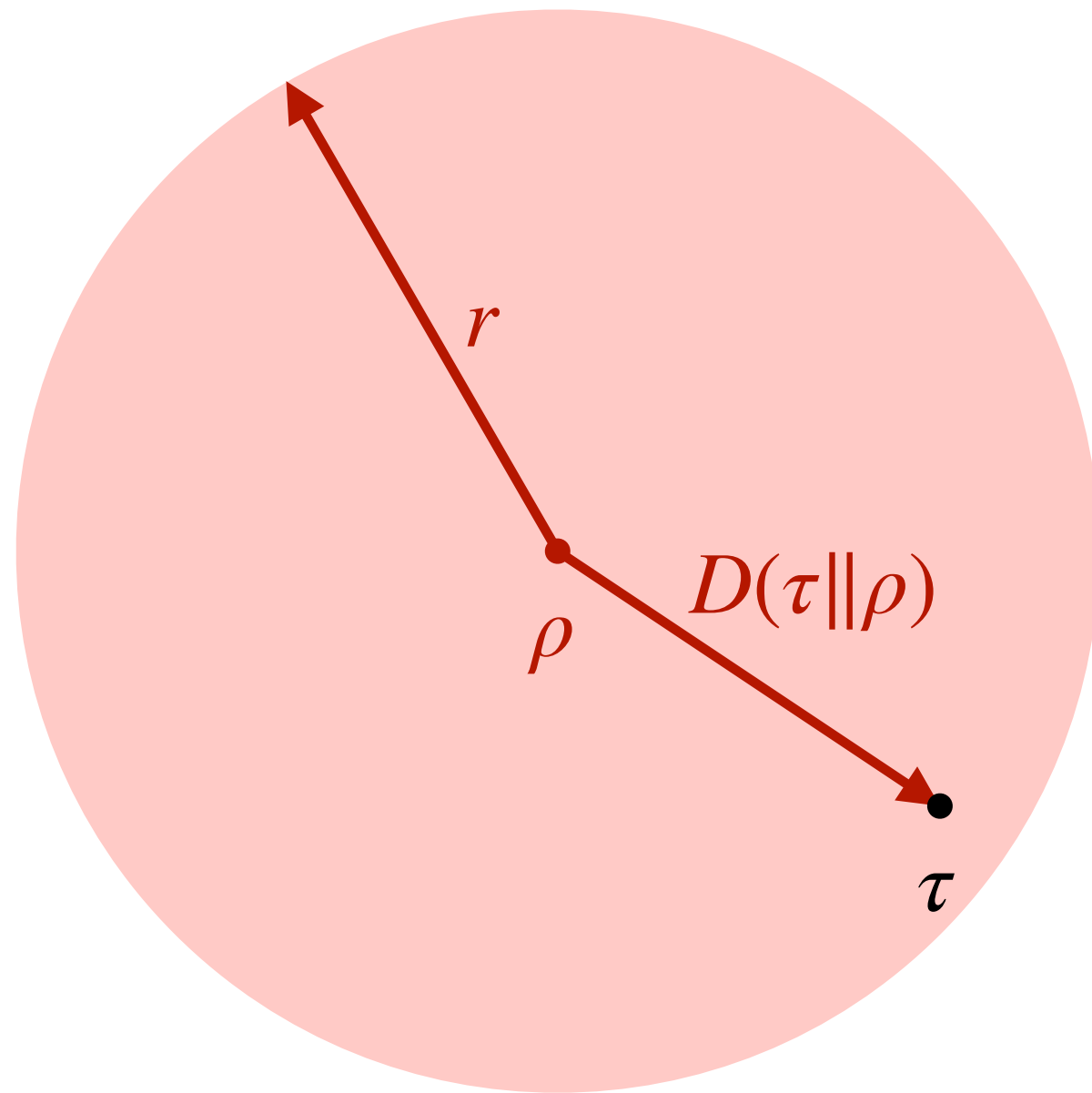
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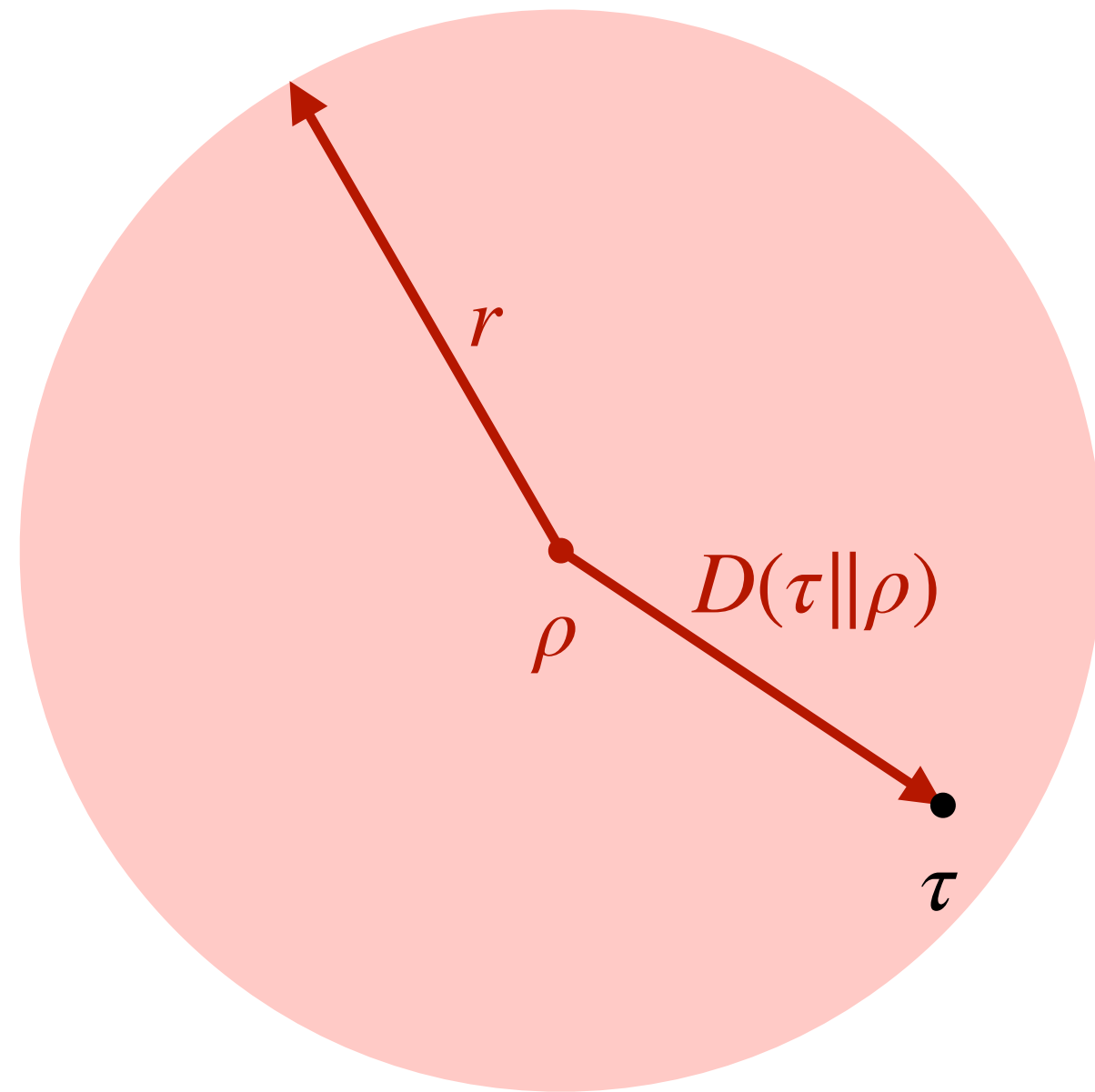
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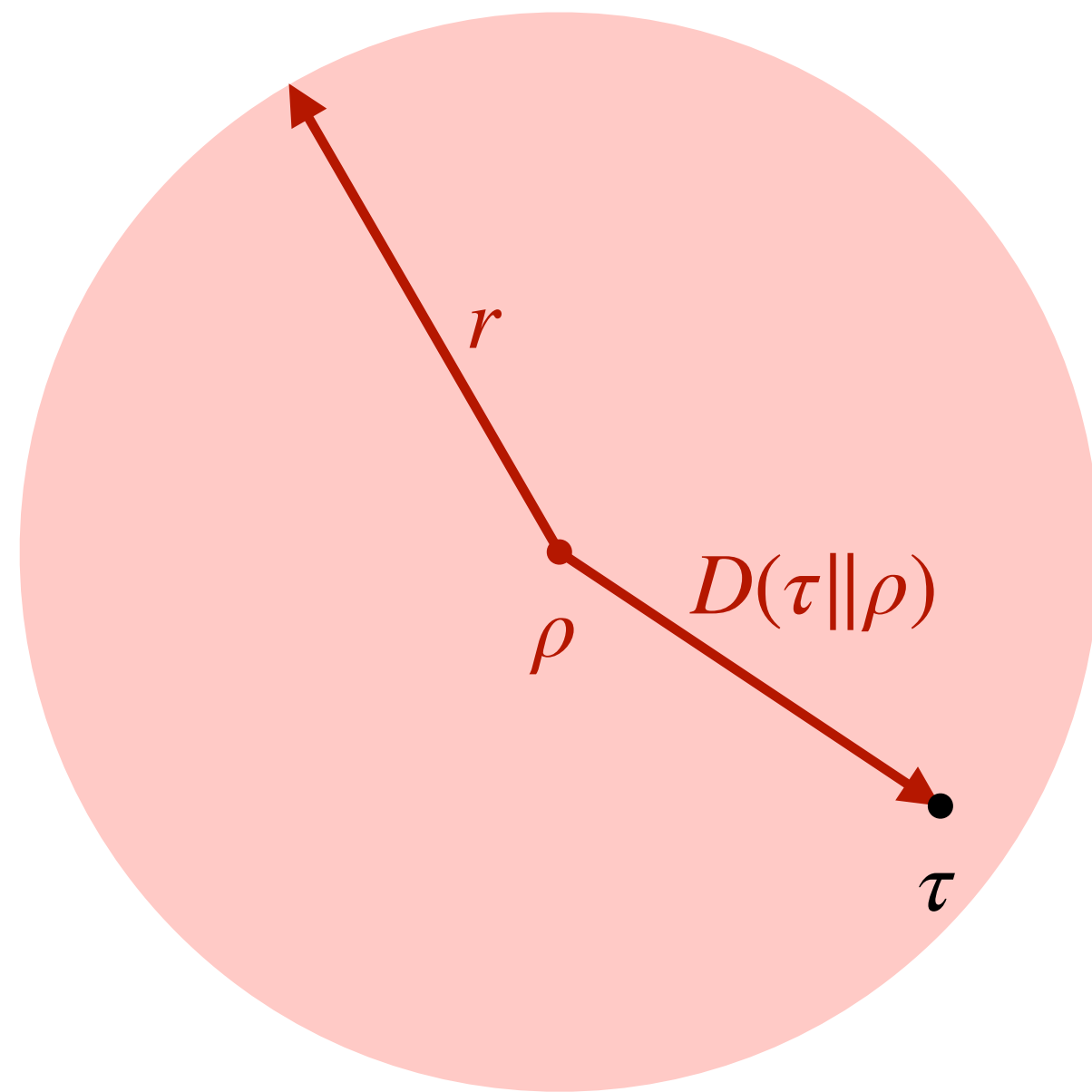
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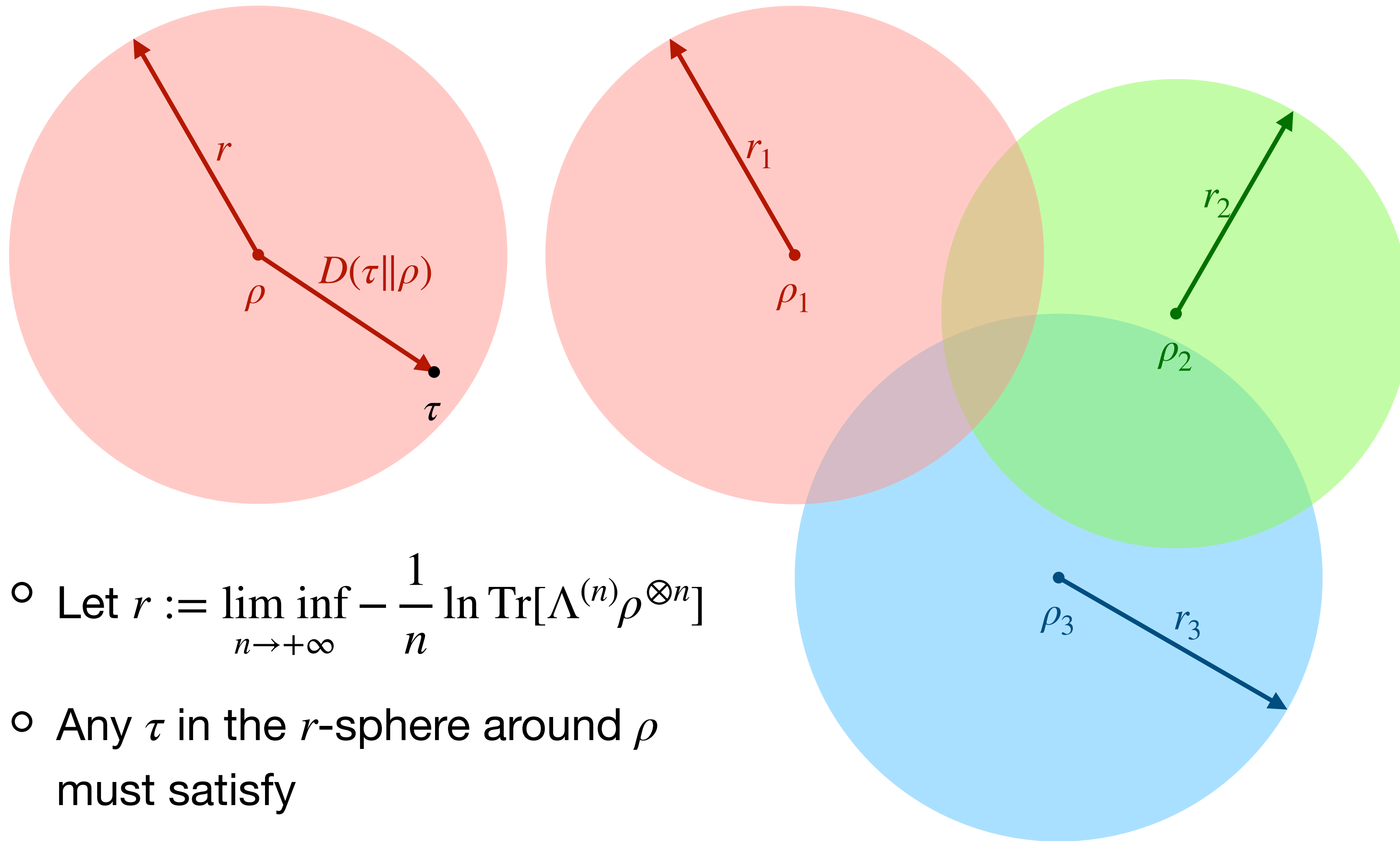


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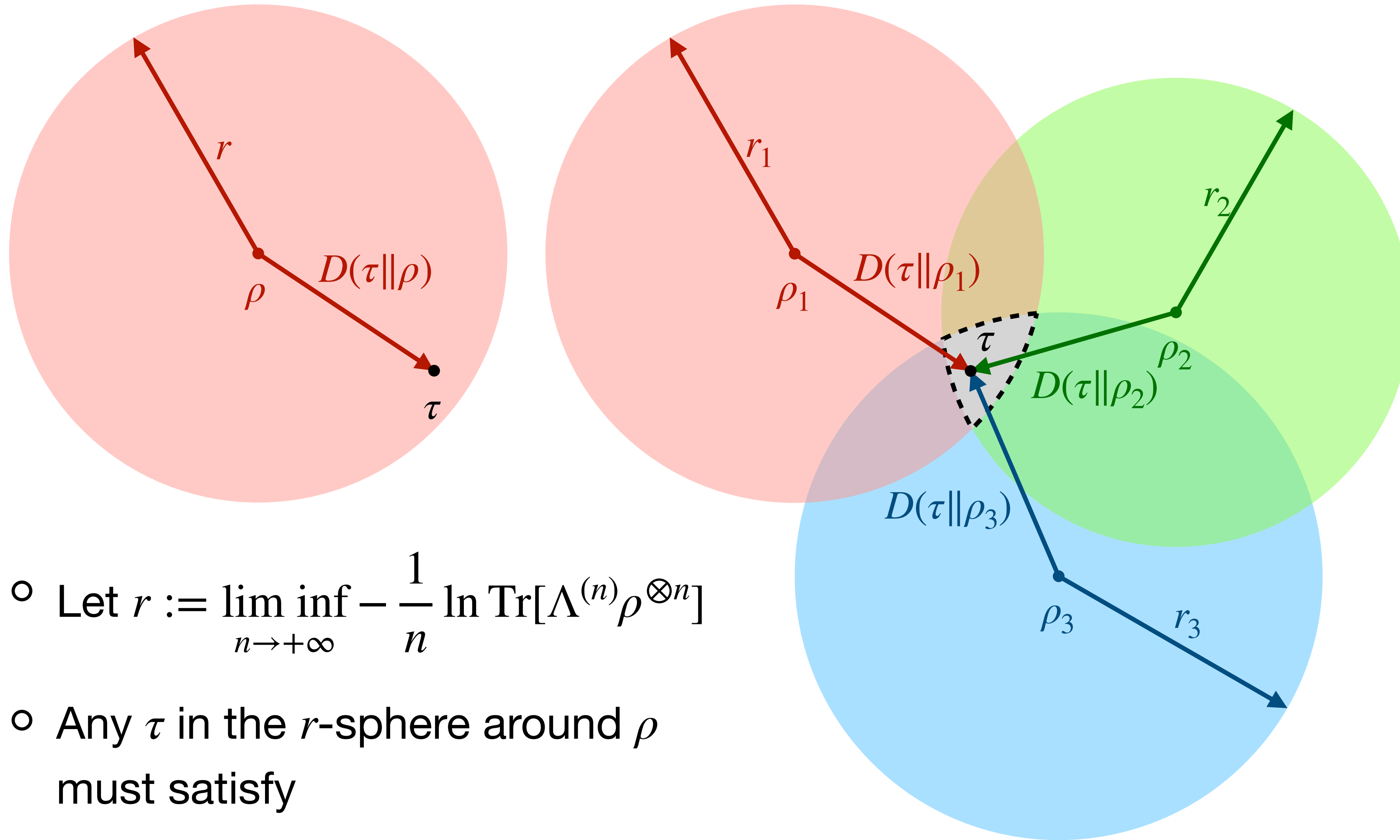
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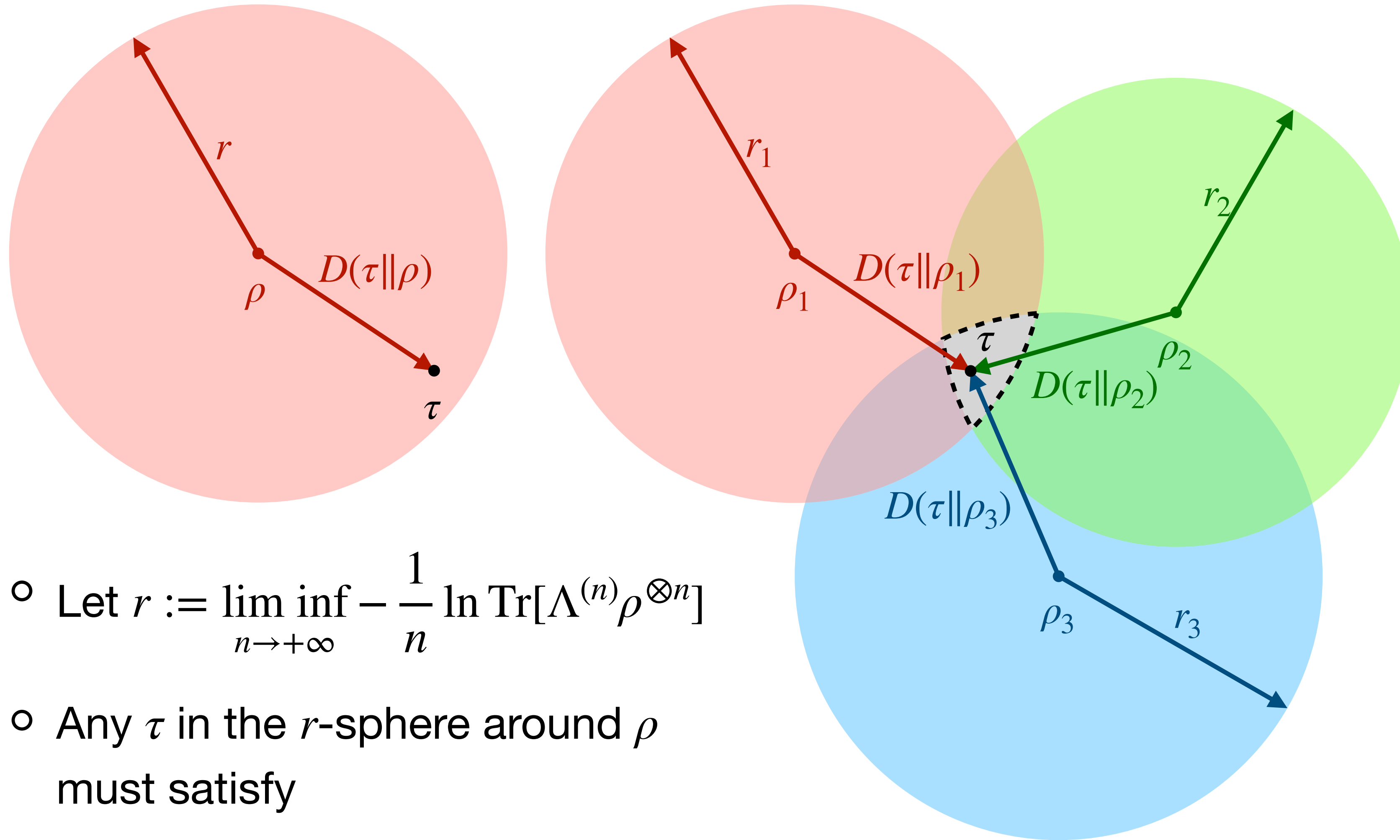
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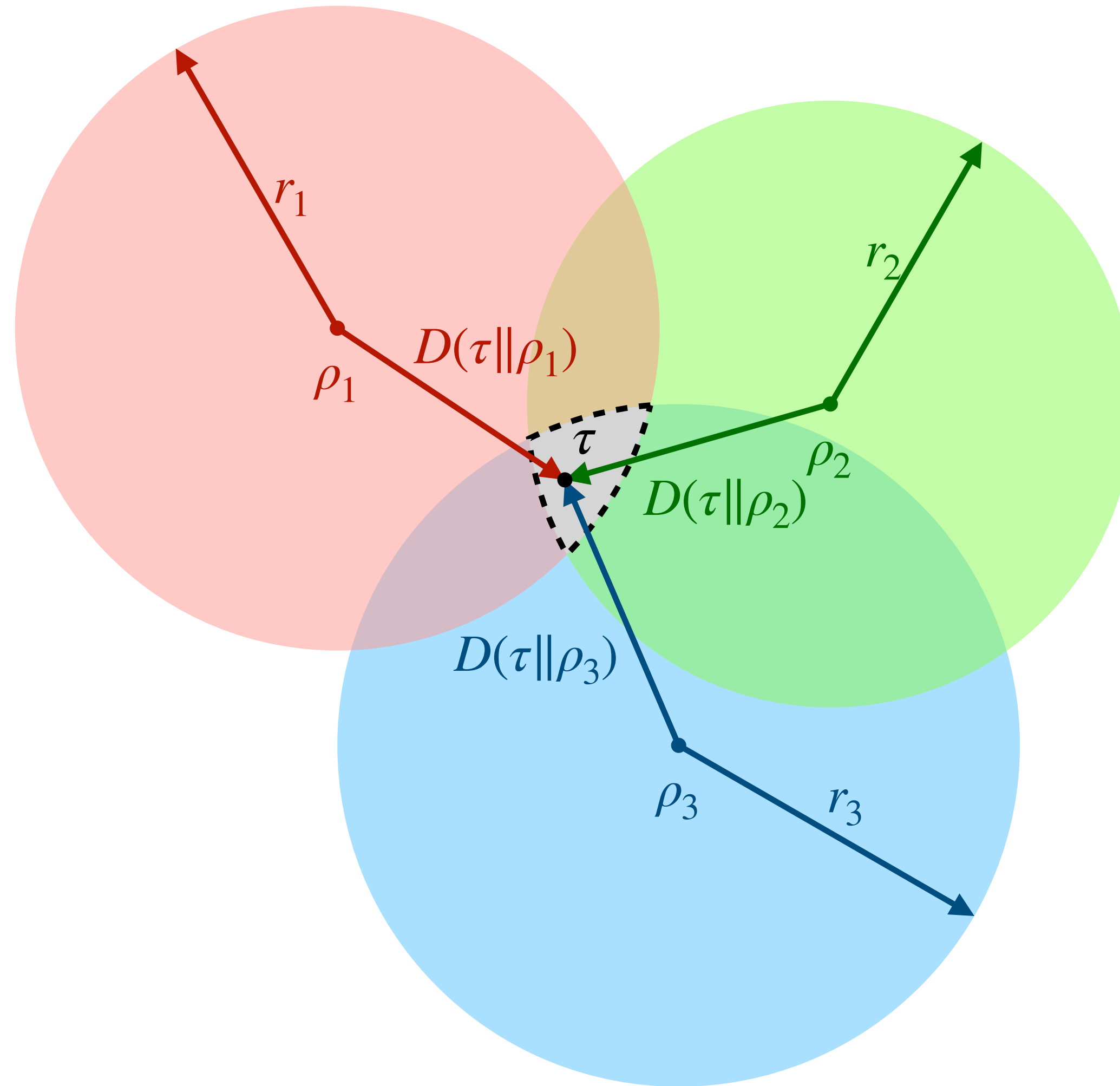
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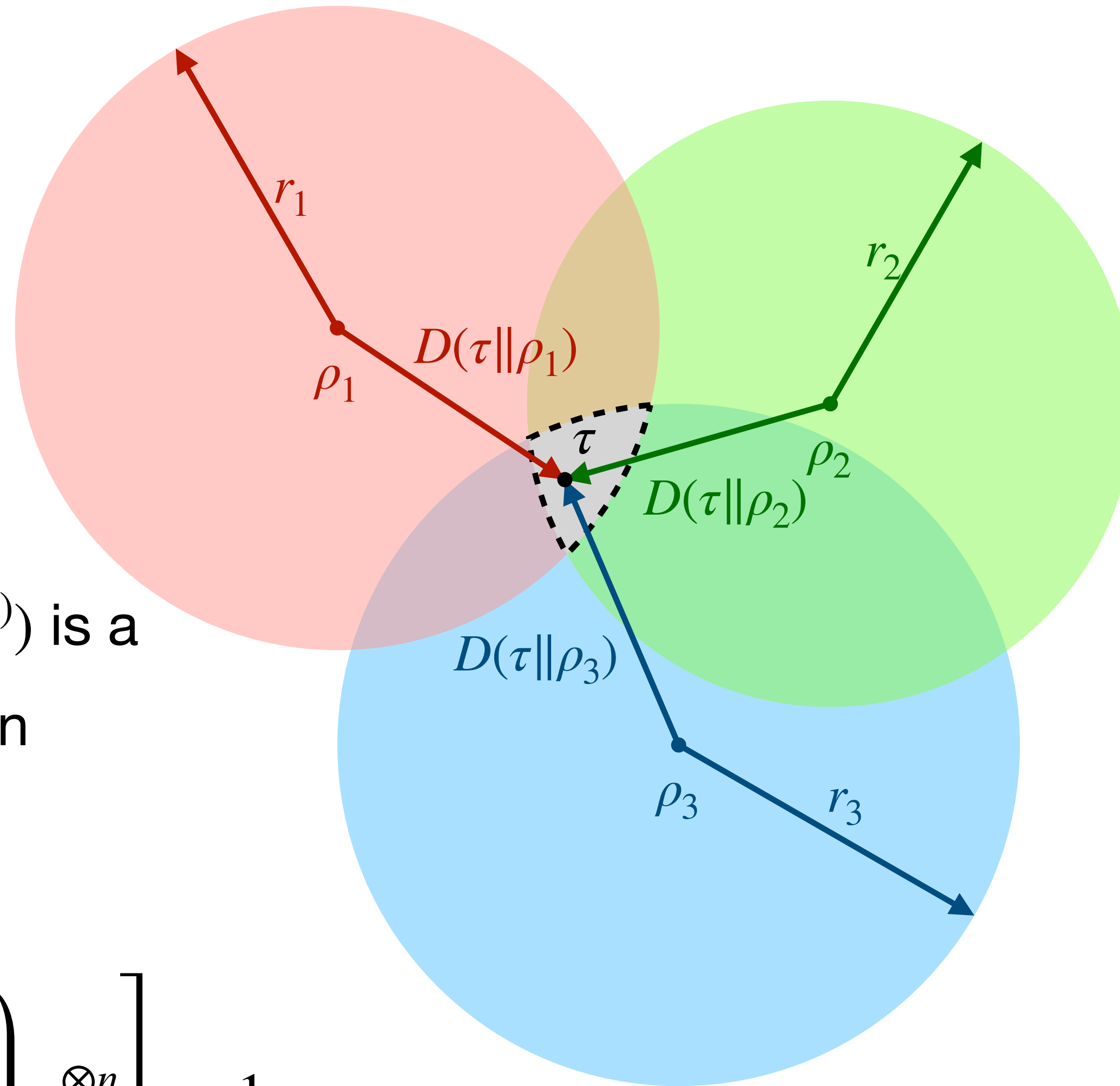
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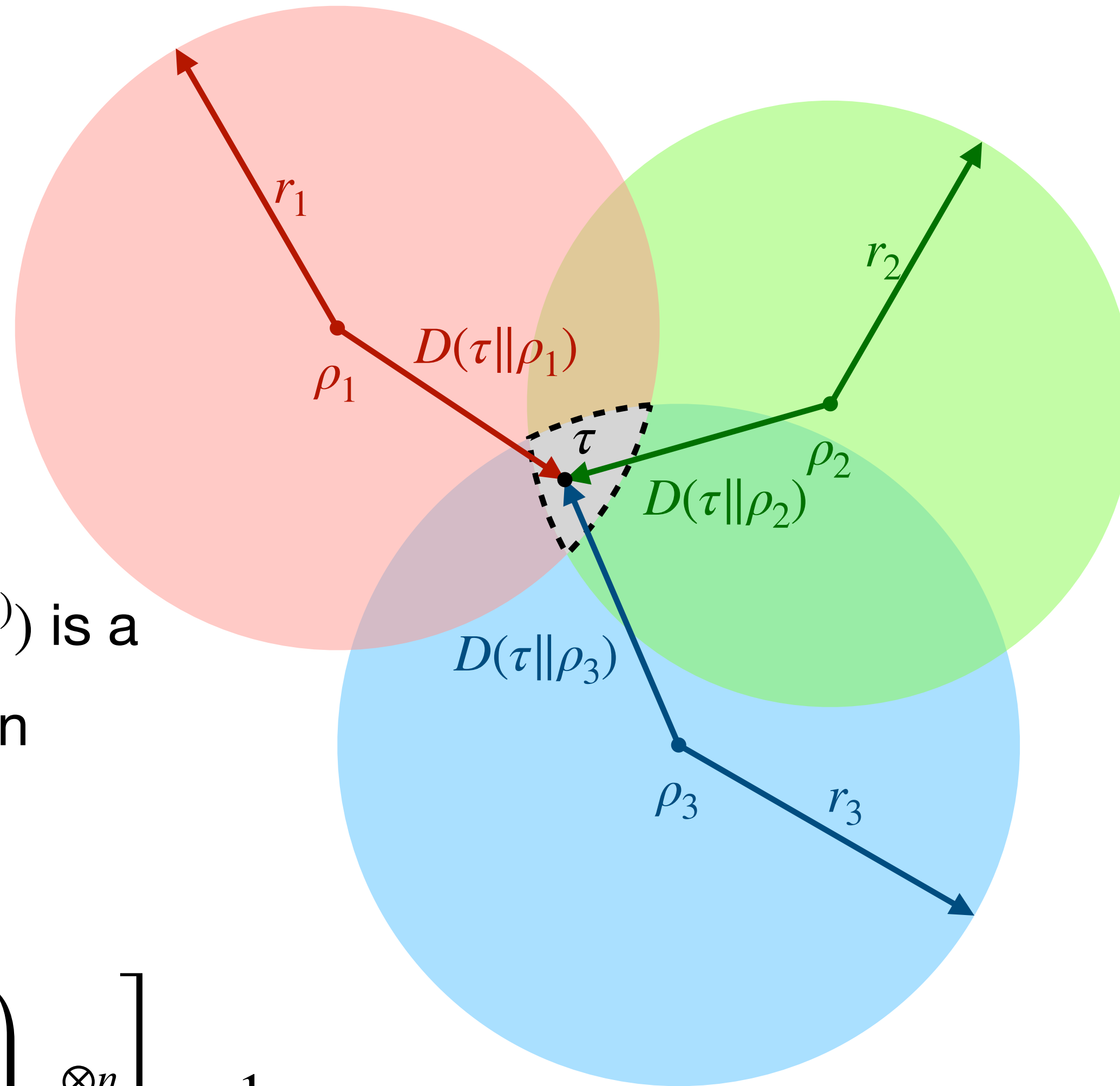
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Contradiction!

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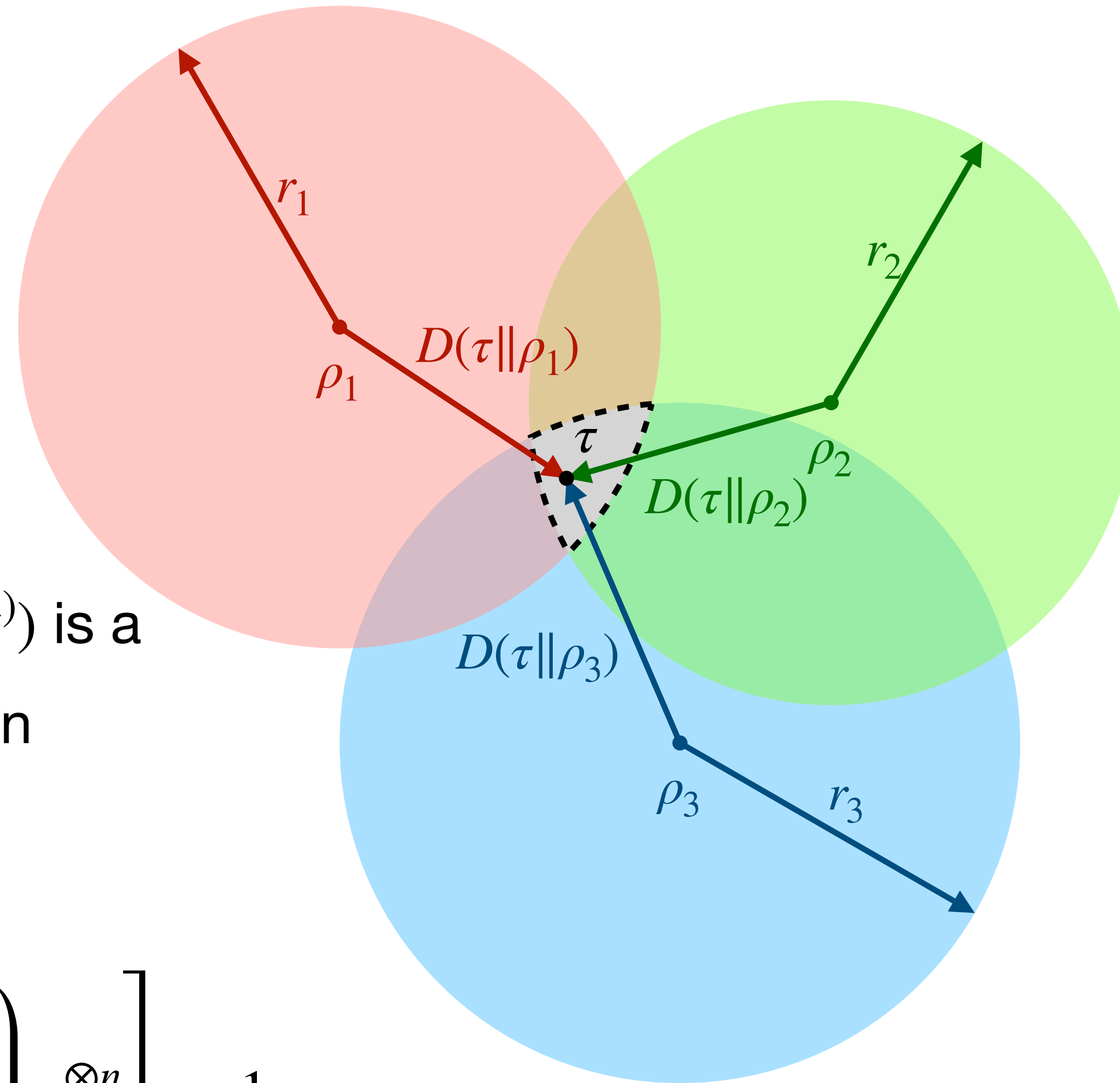
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- Conclusion: r_1, r_2, r_3 has to be such that the intersection of the three spheres is an empty set

- But since $(\Lambda_1^{(n)}, \Lambda_2^{(n)}, \Lambda_3^{(n)})$ is a POVM, by normalisation

$$\sum_{x=1,2,3} \lim_{n \rightarrow +\infty} \text{Tr}[\Lambda_x^{(n)} \tau^{\otimes n}]$$

$$= \lim_{n \rightarrow +\infty} \text{Tr} \left[\left(\sum_{x=1,2,3} \Lambda_x^{(n)} \right) \tau^{\otimes n} \right] = 1$$



Contradiction!

- Consider states $\tau, \rho_1, \rho_2, \rho_3$, and an infinite sequence of POVMs $((\Lambda_1^{(n)}, \Lambda_2^{(n)}, \Lambda_3^{(n)}))_n$
- For $x = 1, 2, 3$, let
$$r_x = \liminf_{n \rightarrow +\infty} -\frac{1}{n} \ln \text{Tr}[\Lambda_x^{(n)} \rho_x^{\otimes n}]$$
- Any τ in the intersection of the three spheres must satisfy

$$\lim_{n \rightarrow +\infty} \text{Tr}[\Lambda_x^{(n)} \tau^{\otimes n}] = 0$$

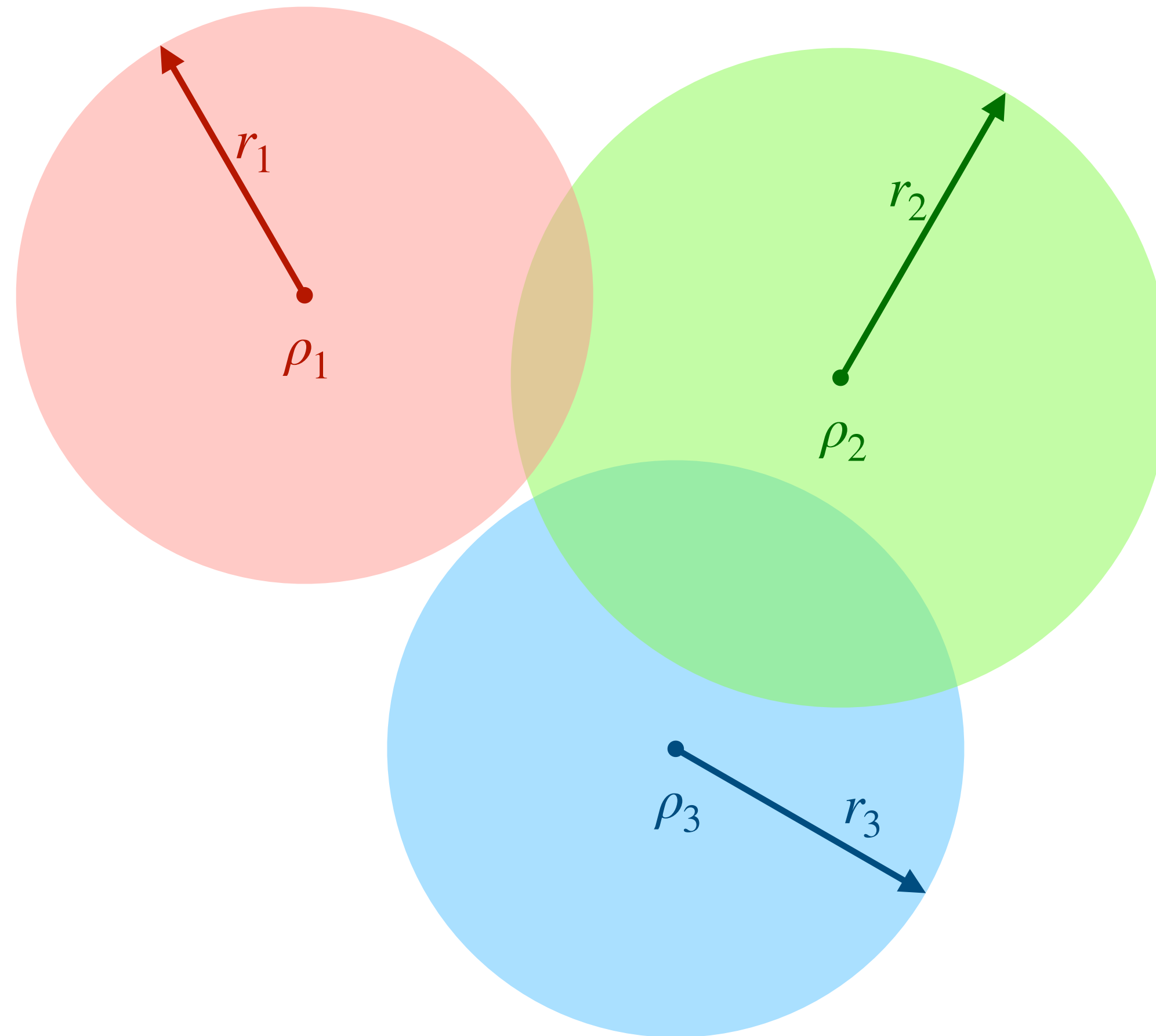
for all $x = 1, 2, 3$

- Implies that

$$\sum_{x=1,2,3} \lim_{n \rightarrow +\infty} \text{Tr}[\Lambda_x^{(n)} \tau^{\otimes n}] = 0$$

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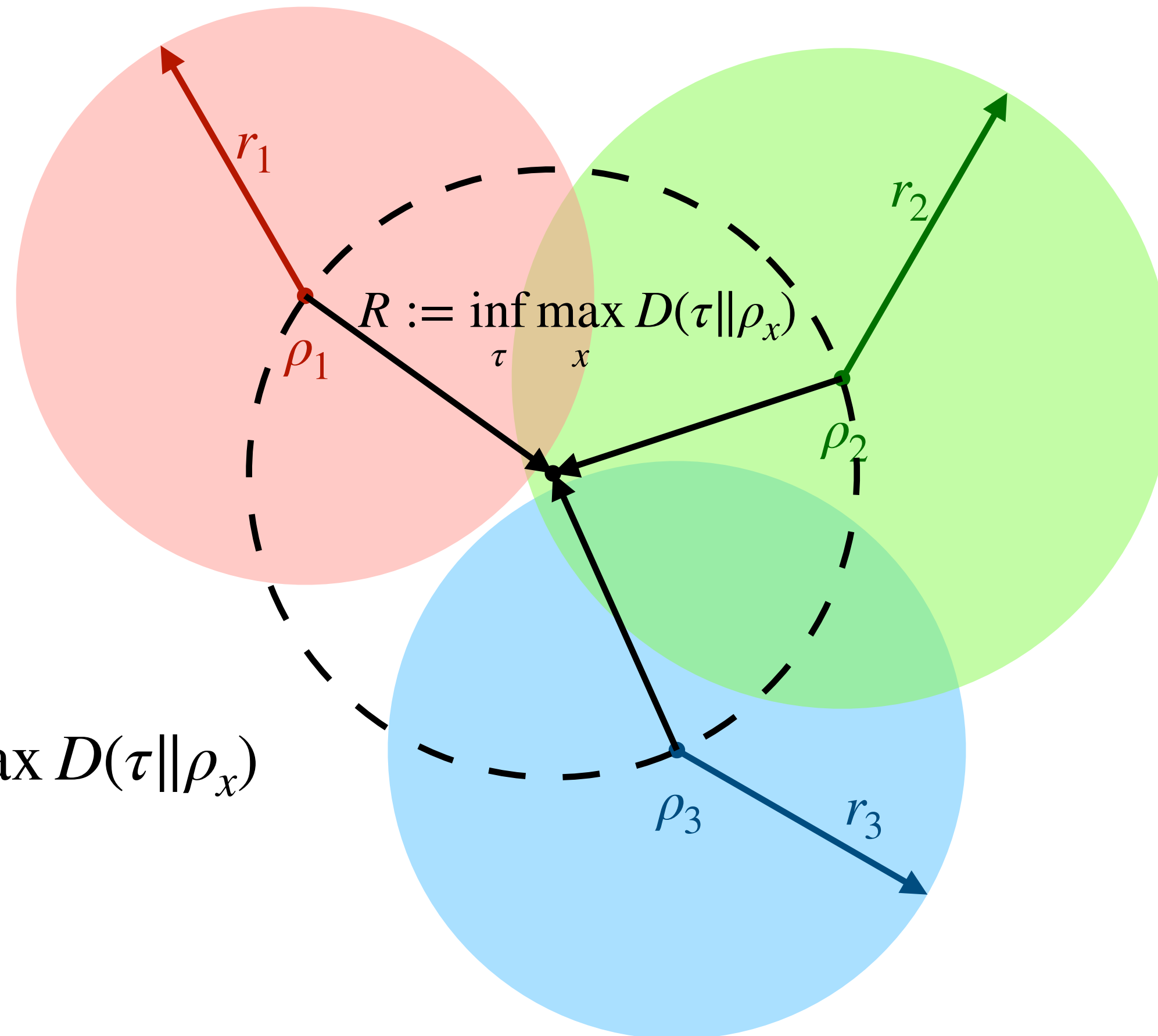
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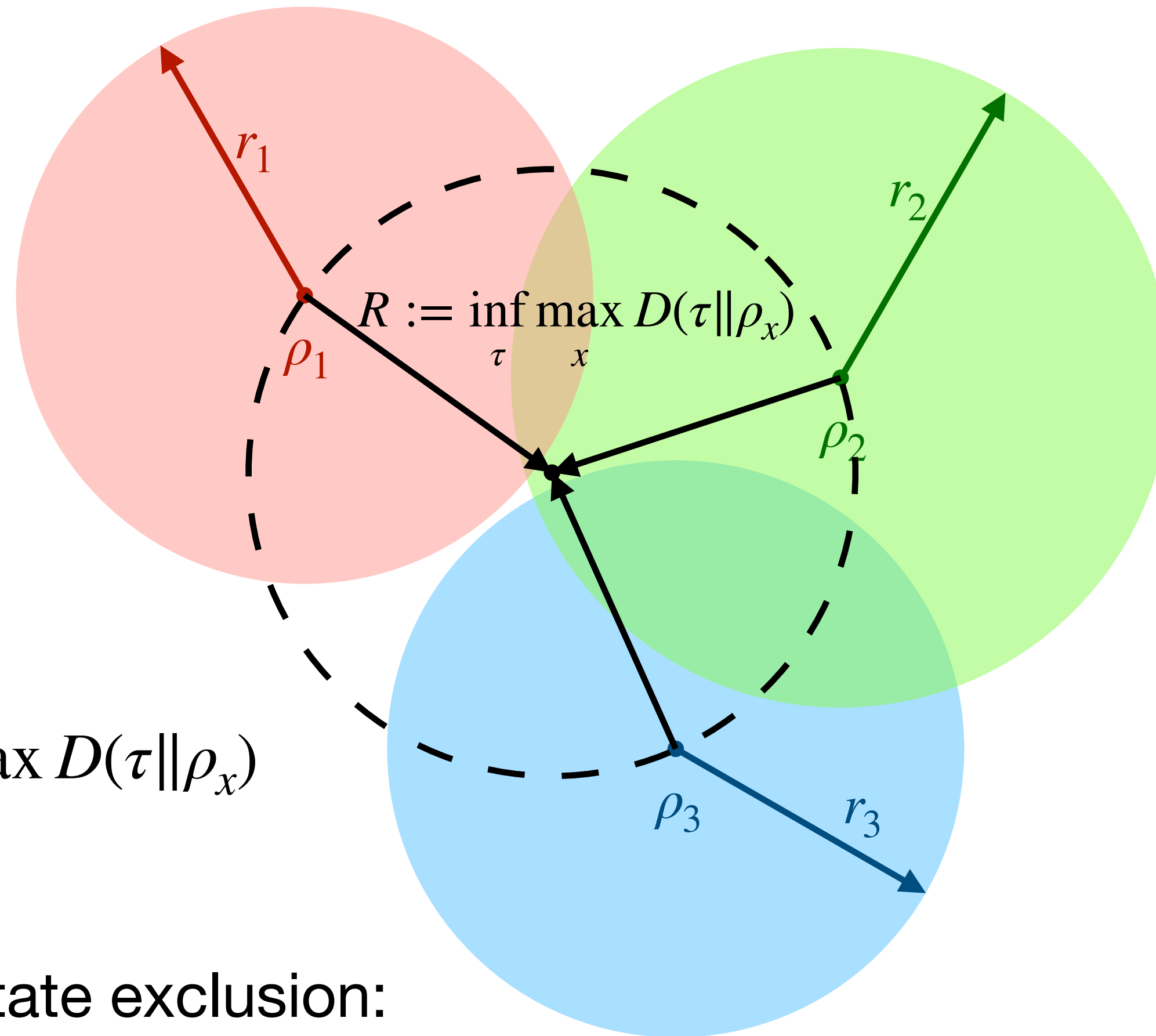
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- Implication in quantum state exclusion:

$$E_{\text{err}}(\mathcal{E}) = \liminf_{n \rightarrow +\infty} -\frac{1}{n} \ln \sum_{x=1,2,3} p_x \text{Tr}[\Lambda_x^{(n)} \rho_x^{\otimes n}] \leq \min\{r_1, r_2, r_3\} \leq \inf_{\tau} \max_x D(\tau \| \rho_x) = C^b((\rho_1, \rho_2, \rho_3))$$



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Quantum Stein's lemma, strong converse, one-shot:

Let τ and ρ be two quantum states and Λ be a measurement operator, with $0 \leq \Lambda \leq I$. Then for $\alpha > 1$,

$$-\ln \text{Tr}[\Lambda\tau] \geq \frac{\alpha - 1}{\alpha} \left(-\ln \text{Tr}[\Lambda\rho] - \widetilde{D}_\alpha(\tau\|\rho) \right)$$

(Mosonyi and Ogawa, 2015)

$$\widetilde{D}_\alpha(\tau\|\rho) := \frac{1}{\alpha - 1} \ln \left\| \rho^{\frac{1-\alpha}{2\alpha}} \tau \rho^{\frac{1-\alpha}{2\alpha}} \right\|_\alpha^\alpha$$

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Quantum Stein's lemma, strong converse, one-shot, extended:

Let τ be a **trace-1 Hermitian operator**, ρ be a quantum state, and Λ be a measurement operator, with $0 \leq \Lambda \leq I$. Then for $\alpha > 1$,

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Theorem: For $\mathcal{E} \equiv ((p_x, \rho_x))_{x \in \mathcal{X}}$ and $\alpha > 1$,

$$-\ln P_{\text{err}}(\mathcal{E}) \leq \sup_{(s_x)_{x \in \mathcal{X}} \in \mathbb{P}} \inf_{\tau \in \text{aff}(\mathcal{D})} \sum_{x \in \mathcal{X}} s_x \widetilde{D}_\alpha(\tau\|\rho_x) + \frac{\alpha}{\alpha - 1} \ln \frac{1}{p_{\min}}$$

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